Genotype and Phenotype Lab

Week Two

1. Purpose and Objectives

The purpose of this investigation is to determine your own genotype for the taste receptor gene TAS2R38 and correlate the genotype with the phenotype, which is the ability to taste phenylthiocarbamide (PTC).

After completing this lab, you should be able to:

- Understand the difference between phenotype and genotype.
- Describe how DNA (deoxyribonucleic acid) sequence (genotype) determines an organism's characteristics (phenotype)
- Observe how a change in DNA sequence may alter phenotype
- Explain how polymerase chain reaction (PCR) can be used to produce copies of a selected segment of DNA
- Explain how restriction fragment length polymorphism can identify differences in genotype
- Describe how DNA gel electrophoresis works.
- Use a bioinformatics resource tool to obtain and explore gene sequences.

2. Safety Guidelines

- All samples must be disposed of in autoclave bags for autoclaving prior to disposal.
- Handle the pipettmen with care. Do not get any liquid on or into any pipettman.
- Always wear safety goggles when using chemicals in the lab.
- Clean and replace all lab equipment. Clean your table when finished.
- Wash your hands before leaving the lab.

3. Background

Polymerase Chain Reaction (PCR)

Recall that PCR is a procedure used to replicate a specific segment of DNA in a test tube. The following components were included in your PCR reaction:

Template DNA (extracted from cheek cells)

Primers – short segments of DNA that will bind to a matching sequence in the TAS2R38 gene

Taq Polymerase – a heat-resistant enzyme that replicates DNA

Deoxynucleoside triphosphates (dNTPs) – monomer subunits for building new DNA strands

The addition of the primers to the cheek cell DNA causes the polymerase to copy only a specific section of the DNA in the samples. For this experiment, the sequence of nucleotides in the primers is designed to match the sequence of nucleotides in part of the taste receptor gene, TAS2R38. Following the PCR run your sample should contain many copies of a short (~200 base pair) segment of this gene.

Restriction Enzyme Digestion

A valuable tool in the molecular biology laboratory is the restriction endonuclease, or restriction enzyme. Restriction enzymes are produced by bacteria to attack foreign DNA that may enter their cells, for example from a virus. These enzymes cut DNA at specific nucleotide sequences. They can be used as a relatively easy way to determine whether a specific sequence is present in DNA and to distinguish between similar but non identical segments of DNA. The are also used to cut pieces of DNA to be used to combine DNA from different sources and create "recombinant DNA."

For this experiment, the enzyme used is called HaeIII. This enzyme will cut DNA at a sequence found in the PCR product from the Taster allele of the TAS2R38 but missing from the PCR product made from the Nontaster allele because of a SNP found in the PCR product.

DNA Gel Electrophoresis

Gel electrophoresis is often used in the molecular biology laboratory to separate segments of DNA based on their size. The "gel" contains agarose, a polysaccharide derived from the cell walls of Red Algae. Agarose is a liquid when heated and will solidify to form a solid material with tiny holes and channels throughout it.

Because DNA is negatively charged at slightly basic pH, it can be induced to move through the matrix of the agarose gel by applying an electrical field. DNA will move toward the positive end of the gel. How quickly DNA segments move through the gel depends on their size. Small DNA fragments can be expected to move quickly through the gel whereas longer fragments will be held up trying to pass through the gel matrix and will move more slowly.

4. Week Two: Restriction Enzyme Digest and Gel Electrophoresis

PreLab

View the Electrophoresis animation on the Cold Spring Harbor DNA Learning Site:

https://dnalc.cshl.edu/resources/animations/gelelectrophoresis.html

Prelab Questions:

1. Your cheek cells contain your entire genome – all the genes that you inherited along with large stretches of DNA that do not contain information for making RNA and proteins. The PCR reaction that you set up last week only makes copies of a small part of this DNA. Part of what gene was copied in the PCR reaction?

2. What determines where a restriction enzyme will cut DNA?

3. Gel electrophoresis separates DNA segments based on their _____.

4. Is DNA negatively or positively charged?

5. Will DNA move toward the positive or negative end of the gel when electricity is applied?

Procedure HaeIII Digestion of PCR Products

1. Obtain your PCR tube and follow the procedure on p. S-8 of the student guide to digest part of your sample with the restriction enzyme.

2. Store the undigested tube and the digested tube on ice.

Procedure Gel Electrophoresis

1. One gel will be poured per lab table.
2. 2 % agarose solution in gel buffer has already been prepared using the following procedure: the agarose powder was added to the buffer in an Erlenmeyer flask and boiled until dissolved. A small amount (0.1 mL/L) of the gelgreen stain was added to the agarose. The agar tube is being kept at 60°C to keep the agarose from solidifying.
3. Assemble the gel caster as instructed.
4. Pour the agarose solution into the gel caster and allow it to solidify. Your gel is ready when it looks milky-white and feels like jello.
5. While waiting for the gel to solidify, calculate what mass of agarose you would need to add to 30 mL of buffer to make a 2 % (w/v) agarose solution. Remember that a 1% solution contains 1 g per 100 mL. **Record this calculation in your lab book.**
6. Carefully pull out the comb from the gel and disassemble the gel caster. Pulling too quickly or pulling sideways may tear the wells, causing your DNA sample to leak out later.
7. Place the gel in the gel tank and make sure it is submerged completely in the running buffer. The wells should fill up with buffer as well. At this point the gel is ready to load.
8. **Write down the order in which samples will be loaded.** Well 1 (left to right) will contain the DNA ladder. See the diagram on p. S-9.
9. Load 5 µL of the 100-bp DNA ladder into the leftmost well of the gel.
10. Load 10 µL of the undigested PCR product for one student in well 2.
11. Load 16 µL of the HaeIII-digested PCR product from the same student in the next lane.
12. Repeat until all students' samples are loaded. Note that the DNA ladder is ***only*** loaded in one lane, so you do not need to add it to any additional wells.
13. Run the gel. This will take 10-30 minutes. Work on the lab questions while you are waiting!
14. Place the transilluminator under the gel and take a photo under blue light.
15. Either print out a copy of your gel to turn in with your lab or upload the photo to Canvas. **Label the lanes and the base pair sizes of the bands in the DNA ladder.**

16. The DNA bp ladder used should have bands at 1500, 1000, 900, 800, 700, 600, 500, 400, 300, 200, and 100 base pairs in size. Some of the larger bands may run together since they will not have been able to migrate far in the gel used.

Questions to be turned in next week:

1. What is the purpose of the GelGreen stain? To what does it bind?
2. Do you see bands on your gel? How many bands are there?
3. Approximately what size are the bands in **each lane** (compare to the size markers- DNA ladder)?